ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C4 · FORCES AND MOTION

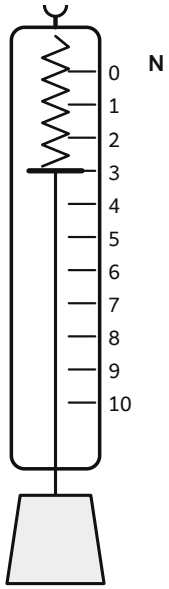
Forces around us

★☆☆
Grade 4

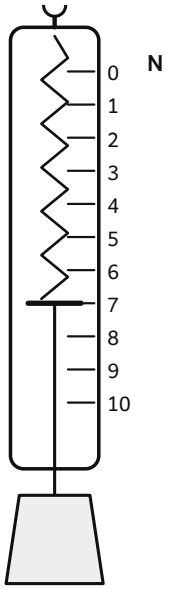
Learning goals

- Name forces.
- Use units.
- Show forces with arrows.

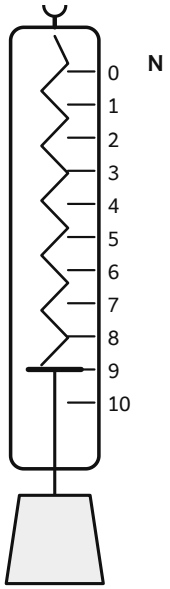
1 Read the forcemeters. [3]



A



B



C

A	B	C
3 N	7 N	9 N

2 Name the force. [5]

What happens	Force
a mango falls from a tree	gravity
a bicycle slows down when you brake	friction
a rope in tug-of-war is pulled tight	tension
a parachute falls slowly	air resistance
a boat floats	upthrust

3 Sort the forces. [5]

SORT THESE

friction **a kick** **gravity** **tension** **magnetic force**

Contact force	Non-contact force
friction	gravity
a kick	magnetic force
tension	

4 Fill in the gaps. [4]

WORD BANK

smaller **newtons** **bigger** **metres** **kilograms** **same**

Mass is measured in kilograms. Weight is a force, so it is measured in newtons.

On the Moon your mass stays the same but your weight is smaller.

ANSWER KEY

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THINK IT THROUGH · C4 · FORCES AND MOTION

Forces in action

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Learning goals

- Draw force arrows.
- Explain motion using the laws of motion.

1 **Draw arrows to show the forces on a boy on a slide: gravity pulling him down and friction slowing him.** [2]

Slide

Arrow pointing down (weight/gravity); arrow pointing up the slide, opposite to the movement (friction).

2 **Explain each one using a law of motion.** [6]

What happens	Explanation
a football rolling on grass slowly stops	law 1: friction (a force) acts on it and stops it
it is harder to push a full rickshaw van than an empty one	law 2: more mass needs a bigger force to change its speed
a swimmer pushes water back and moves forward	law 3: the water pushes the swimmer forward

3 **Why do people put sand or gravel on a slippery muddy path in the monsoon?** [2]

Sand increases friction so feet and tyres grip and do not slip.

4 **In carrom, what happens to the striker's movement when it hits a coin?** [2]

The striker slows down and passes some of its movement (kinetic) energy to the coin, which starts to move.

ANSWER KEY

INVESTIGATE · C4 · FORCES AND MOTION

Which surface has the most friction?

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Learning goals

- Plan a fair test.
- Measure distance.
- Explain results using friction.

Class 4D let a toy car roll down the same ramp onto four different surfaces and measured how far it travelled before stopping.

1 **Complete the plan.** [3]

Change	the surface
Measure	distance the car travels (cm)
Keep the same	the car, the ramp height and slope, where the car starts, no push

2 **Here are the results (mean of three tries). Draw a bar chart.** [3]

Surface	tiles	wood	jute mat	sand paper
Distance (cm)	180	140	70	35



3 **Which surface had the most friction? How do you know?** [2]

Sandpaper: the car travelled the shortest distance, so friction slowed it most.

4 **Why did they do each test three times and find the mean?** [1]

To make the results more reliable.

5 **Which surface would be best for a playground slide landing area? Why?** [1]

A rough surface with lots of friction (like a jute mat) so children stop safely.

ANSWER KEY

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LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C4 · FORCES AND MOTION

Kabaddi and tug-of-war: the forces in our games

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Learning goals

- Find forces in real games.
- Use data from a tug-of-war.
- Think about fairness and safety.

Science | Physical education | Maths | Culture

Kabaddi is the national game of Bangladesh. A raider runs into the other team's half, tries to tag players and get back, while defenders try to hold them. Players **push**, **pull**, twist and change direction quickly. They play barefoot on soil or a mat, which gives good **friction** for gripping the ground.

In **tug-of-war** (*dori tana*), two teams pull a rope in opposite directions. The rope is in **tension**. If both teams pull with the same force, the forces are **balanced** and the rope does not move. If one team pulls harder, the forces are **unbalanced** and the rope moves towards the stronger team.

Example game data below.

1 **Two teams pull a rope (example data).** [4]

Round	Team Shapla (N)	Team Doel (N)	Which way does the rope move?
1	800	800	it does not move (balanced)
2	900	750	towards Shapla
3	700	850	towards Doel

a) What is the difference in force in round 2?

900 − 750 = 150 N towards Shapla

2 **Why do kabaddi players play barefoot on soil or a mat, not on smooth tiles?** [2]

Soil and mats have more friction, so their feet grip and they can push off, stop and turn without slipping.

3 **When a raider pushes off the ground to run, which law of motion is at work? Explain.** [2]

Law 3: the feet push back on the ground and the ground pushes the raider forward.

4 **Suggest two rules that make kabaddi or tug-of-war safe and fair.** [2]

e.g. teams of the same total mass; no pulling hair or clothes; a soft surface; gloves for the rope; a referee.

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REVIEW AND CHALLENGE · C4 · FORCES AND MOTION

Show what you know

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Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

J	T	W	M	O	U	Y	H	F	H
I	V	E	A	F	M	V	J	O	C
C	L	I	S	R	O	J	G	G	L
Y	J	G	S	I	T	T	U	R	F
E	P	H	F	C	I	A	P	A	F
F	I	T	P	T	O	F	U	V	O
C	S	R	U	I	N	E	L	I	R
V	J	D	S	O	T	K	L	T	C
J	L	C	H	N	A	B	Y	Y	E
N	E	W	T	O	N	I	J	H	P

force friction gravity mass motion newton pull push weight

2 Which is a non-contact force? [1]

- ○ friction
- ○ tension
- ● **gravity**
- ○ a push

3 What unit is used for weight? [1]

- ○ kilograms
- ● **newtons**
- ○ metres
- ○ litres

4 Draw a ball being kicked. Use arrows to show the kick, gravity and air resistance. [3]

Kicked ball

Kick arrow in the direction of movement; gravity arrow down; air resistance arrow opposite to the movement.

5 **Challenge. An astronaut has a mass of 60 kg. What is her mass on the Moon? Is her weight bigger or smaller there? Explain.** [2]

Her mass is still 60 kg (the amount of matter does not change). Her weight is smaller (about 6 times less) because the Moon's gravity is weaker.